

Counterfactual Group-Aware Debiasing for Fair Ad Ranking under Exposure Confounding

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Abstract—Learning to rank from biased click feedback is a fundamental challenge in computational advertising. Existing unbiased learning-to-rank (ULTR) methods address position bias through inverse propensity scoring or doubly robust estimation, yet they overlook systematic exposure disparities across ad groups defined by attributes such as topic category or cold-start status. We propose Counterfactual Group-Aware Debiasing (CGAD), a two-layer framework that jointly corrects position bias and group-level exposure confounding. The first layer estimates group-conditioned examination propensities, and the second layer trains a ranker via group-aware counterfactual risk minimization augmented with a fairness regularizer. We further introduce the Group-Conditioned Counterfactual Exposure Gap (GCCEG), a metric that quantifies deviation from merit-proportional exposure across groups after counterfactual correction. Experiments on the Criteo and Avazu advertising datasets with semi-synthetic bias injection demonstrate that CGAD maintains competitive ranking accuracy while reducing cross-group exposure disparity and improving worst-group performance.

Index Terms—Unbiased learning to rank, counterfactual inference, fairness of exposure, ad ranking, group-aware debiasing

I. INTRODUCTION

Click-through rate prediction and ad ranking systems learn predominantly from user interaction logs, which are inherently biased. Users examine items at higher positions with greater probability, a phenomenon known as position bias [1]. Unbiased learning-to-rank (ULTR) methods aim to correct this bias so that rankers trained on biased click data can recover the true relevance ordering. The dominant paradigm uses inverse propensity scoring (IPS) [2] or its extensions such as the dual learning algorithm (DLA) [3] and doubly robust (DR) estimators [4].

However, real-world ranking systems encounter biases more complex than position effects [5]. In advertising, the logging policy is shaped by prior relevance estimates and advertiser budgets, creating confounding between ad characteristics and exposure. Ads from different groups—defined by topic category, brand maturity, or cold-start status—receive systematically different exposure, and under-exposed ads accumulate fewer clicks, reinforcing a feedback loop.

The fair ranking literature has established that utility-maximizing rankers can produce persistent exposure imbalances across item groups [6]. Fairness-of-exposure constraints require that groups receive exposure commensurate with their merit [7], yet these interventions are typically applied as post-processing and do not interact with counterfactual debiasing. We argue that

position debiasing and group-fair exposure allocation should be addressed jointly: IPS-corrected rankers may still under-expose certain groups, and fairness regularizers applied to biased rankers may redistribute exposure based on biased relevance estimates.

We propose **Counterfactual Group-Aware Debiasing (CGAD)**, a framework that unifies counterfactual debiasing and group-aware fairness. Our contributions are:

- A two-layer framework that estimates group-conditioned propensities and trains a ranker through group-aware counterfactual risk minimization with integrated fairness constraints.
- A new evaluation metric, the *Group-Conditioned Counterfactual Exposure Gap* (GCCEG), which measures deviation between the exposure each group receives and the exposure its relevance warrants.
- Empirical evaluation on Criteo and Avazu with semi-synthetic bias injection, showing that CGAD achieves competitive accuracy while improving exposure fairness and worst-group performance.

II. METHODOLOGICAL FOUNDATION

Counterfactual learning has emerged as a fundamental methodological paradigm for ranking optimization under observational feedback. Rather than treating user interactions as direct reflections of relevance, this paradigm formulates ranking as a causal estimation problem in which observed clicks are jointly determined by latent relevance and exposure processes. Within this perspective, counterfactual risk minimization establishes a statistically grounded mechanism for recovering unbiased relevance information from biased interaction logs, while recent advances in robust counterfactual optimization further strengthen estimation reliability under imperfect propensity specification and distributional variability [8]. In parallel, fairness-aware ranking introduces exposure allocation as an integral component of the optimization objective and formalizes the principle that exposure should be distributed according to estimated merit [9]. These developments collectively characterize ranking as a unified causal decision-making process involving both relevance estimation and exposure allocation, thereby providing the theoretical foundation for integrating counterfactual debiasing and group-aware fairness within a common learning framework.

Modeling exposure dynamics at the group level requires learning mechanisms capable of capturing structural

heterogeneity beyond observed interaction frequencies. Methodological advances in self-supervised representation learning demonstrate that latent semantic relationships can be extracted from heterogeneous data through intrinsic supervisory signals, enabling the discovery of invariant structures that remain robust under sparse supervision [10]. Related developments in graph-based relational modeling reveal the effectiveness of representing indirect dependencies and higher-order interactions through structured relational propagation, facilitating the characterization of complex relationships among entities, contextual attributes, and behavioral signals [11]. Complementary progress in meta-learning further establishes principled approaches for transferring knowledge across unevenly represented data distributions and enhancing generalization in data-sparse regions of the feature space [12]. Adaptive federated learning and decentralized aggregation frameworks extend these ideas by introducing mechanisms for learning under heterogeneous local distributions while preserving globally consistent optimization objectives [13], [14]. Collectively, these methodological directions provide a coherent foundation for modeling group-conditioned exposure patterns and motivate the incorporation of heterogeneous-group awareness into the propensity estimation process.

Uncertainty-aware inference constitutes a second foundational principle underlying the proposed framework. Since relevance and exposure processes are only partially observable, counterfactual estimation inherently involves uncertainty regarding both latent outcomes and causal effects. Recent advances in uncertainty-aware attribution modeling and intelligent decision-making demonstrate the value of explicitly incorporating confidence estimation into observational inference procedures, thereby improving the reliability of downstream optimization decisions [15]. Counterfactual reasoning frameworks similarly emphasize the estimation of alternative outcomes under hypothetical interventions, enabling learning systems to approximate unbiased target distributions despite the presence of confounded observations [16]. Risk-calibrated optimization methodologies further extend this perspective by integrating uncertainty-sensitive objectives that account for varying confidence levels across decision contexts and operational conditions [17]. Viewed collectively, these studies establish uncertainty-aware counterfactual estimation as a natural methodological basis for group-aware risk minimization and for evaluating exposure allocation relative to estimated relevance.

A further methodological influence originates from recent developments in adaptive intelligence systems and modular learning architectures. Contemporary research increasingly views complex learning tasks as compositions of multiple interconnected optimization stages, where intermediate estimation modules progressively refine the information available to downstream decision processes. Adaptive reasoning frameworks demonstrate the effectiveness of dynamically adjusting inference strategies according to contextual knowledge availability and task requirements [18]. Cost-aware information acquisition mechanisms similarly illustrate how complementary evidence sources can be integrated through coordinated optimization procedures that balance predictive effectiveness and resource efficiency [19]. Evidence-guided verification frameworks contribute additional methodological insights by emphasizing the importance of validating intermediate inference outcomes before final decision generation [20]. Structured reasoning architectures further reinforce this principle through the incorporation of verification and consistency-checking mechanisms that enhance reliability across multi-stage learning pipelines [21]. Together, these developments provide strong

methodological support for a layered architecture in which propensity estimation, counterfactual correction, and fairness-aware optimization operate as complementary components within a unified learning system. Context-aware optimization represents an additional methodological theme that informs the overall design philosophy of the proposed approach. Recent research on semantic-aware resource management, adaptive scheduling, workload-sensitive optimization, and intelligent routing consistently demonstrates the advantages of conditioning learning behavior on contextual characteristics rather than relying on globally uniform decision policies [22]–[25]. Across these frameworks, optimization objectives are dynamically adjusted according to environmental properties, distributional patterns, and operational constraints, enabling more effective adaptation to heterogeneous conditions. Such a perspective naturally extends to ranking environments, where exposure generation mechanisms may vary across distinct item groups and behavioral contexts. Consequently, group-conditioned propensity estimation can be viewed as a context-aware extension of conventional counterfactual learning, allowing exposure correction to account for heterogeneous exposure dynamics while preserving the causal interpretation of the learning objective.

III. METHODOLOGY

A. Problem Setting

Consider an ad ranking system that, for each query context q , produces a ranked list of n candidate ads $\mathbf{d} = (d_1, \dots, d_n)$. Each ad d_i has latent true relevance $r_i \in \{0, 1\}$ and belongs to a group $g_i \in \mathcal{G} = \{1, \dots, G\}$ determined by a categorical attribute. The system observes click feedback $c_i \in \{0, 1\}$ generated according to:

$$P(c_i = 1 | q, d_i, k_i, g_i) = P(o_i = 1 | k_i, g_i) \cdot P(r_i = 1 | q, d_i), \quad \#(1)$$

where k_i is the position of d_i and o_i denotes examination. This follows the position-based click model (PBM) assumption, which we generalize by allowing the examination probability to depend on the group identity g_i .

B. Layer 1: Group-Conditioned Propensity Estimation

We parameterize the examination probability as:

$$\hat{p}(o = 1 | k, g; \phi) = \sigma(\alpha_k + \beta_g + \delta_{k,g}), \quad \#(2)$$

where $\sigma(\cdot)$ is the sigmoid function, α_k captures the position effect, β_g captures the group-level baseline shift, and $\delta_{k,g}$ is a position-group interaction term. The parameters $\phi = \{\alpha_k, \beta_g, \delta_{k,g}\}$ are estimated by adapting the DLA framework: given click data $\mathcal{D} = \{(q_j, d_j, k_j, g_j, c_j)\}_{j=1}^N$, we minimize:

$$\mathcal{L}_{\text{prop}}(\phi) = - \sum_{j=1}^N \left[c_j \log \hat{p}_j + (1 - c_j) \log (1 - \hat{p}_j) \right], \quad \#(3)$$

where $\hat{p}_j = \hat{p}(o = 1 | k_j, g_j; \phi) \cdot \hat{r}(q_j, d_j; \theta)$ and $\hat{r}$ is the relevance estimate from the current ranker. During the propensity update step, $\hat{r}(q_j, d_j; \theta)$ is fixed and only ϕ is updated. The

ranker parameters θ are then updated in the subsequent step, following an alternating optimization scheme.

C. Layer 2: Group-Aware Counterfactual Risk Minimization

Given the estimated propensities, we define the group-aware counterfactual ranking loss:

$$\mathcal{L}_{\text{GA-CR}}(\theta) = \frac{1}{N} \sum_{j=1}^N \frac{c_j}{\widehat{p}(o = 1|k_j, g_j; \phi)} \cdot \ell(f_\theta(q_j, d_j), c_j), \quad \#(4)$$

where f_θ is the ranker's scoring function and $\ell(\cdot, \cdot)$ is a pointwise loss. The propensity denominator conditions on both position k_j and group g_j , thereby correcting for group-specific examination disparities.

To prevent exposure from diverging across groups of equal relevance, we add a fairness regularizer. Let $\text{Exp}(g|\pi_\theta)$ denote the average exposure that ads in group g receive under ranking policy π_θ :

$$\text{Exp}(g|\pi_\theta) = \frac{1}{|\mathcal{D}_g|} \sum_{j \in \mathcal{D}_g} v(k_j^\theta), \quad \#(5)$$

Algorithm 1 CGAD Training

Require: Click log $\mathcal{D}$, group assignment function $g(\cdot)$, trade-off λ

- 1: Initialize ranker f_θ , propensity model $\widehat{p}_\phi$
 - 2: **for** each epoch **do**
 - 3: **Step 1:** Fix θ ; update ϕ by minimizing $\mathcal{L}_{\text{prop}}$ (Eq. 3)
 - 4: **Step 2:** Compute IPS weights using $\widehat{p}(o|k, g; \phi)$
 - 5: **Step 3:** Fix ϕ ; update θ by minimizing $\mathcal{L}(\theta)$ (Eq. 7)
 - 6: **end for**
 - 7: **return** Trained ranker f_θ
-

where $\mathcal{D}_g$ is the set of instances in group g , k_j^θ is the position assigned by f_θ , and $v(k) = 1/\log_2(k+1)$ is the standard position discount. The Exposure Fairness Gap (EFG) is defined as:

$$\text{EFG}(\theta) = \sum_{g=1}^G \left| \frac{\text{Exp}(g|\pi_\theta)}{\sum_{g'} \text{Exp}(g'|\pi_\theta)} - \frac{\text{Rel}(g)}{\sum_{g'} \text{Rel}(g')} \right|, \quad \#(6)$$

where $\text{Rel}(g)$ denotes the average estimated relevance of group g . This quantity also serves as the fairness regularizer $\Omega_{\text{fair}}(\theta) = \text{EFG}(\theta)$. The overall training objective is:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{GA-CR}}(\theta) + \lambda \cdot \Omega_{\text{fair}}(\theta), \quad \#(7)$$

where $\lambda \geq 0$ controls the accuracy-fairness trade-off. Algorithm 1 summarizes the training procedure.

D. Evaluation Metric: GCCEG

We propose the *Group-Conditioned Counterfactual Exposure Gap* (GCCEG):

$$\text{GCCEG} = \max_{g \in \mathcal{G}} \left| \frac{\text{Exp}(g|\pi_\theta)}{\text{Rel}(g)} - \frac{1}{G} \sum_{g'} \frac{\text{Exp}(g'|\pi_\theta)}{\text{Rel}(g')} \right|, \quad \#(8)$$

While EFG captures aggregate exposure disparity across groups, it may overlook substantial unfairness experienced by individual groups. To address this limitation, GCCEG evaluates the maximum deviation of any group's exposure-to-relevance ratio from the overall average, thereby providing a worst-case fairness assessment. A value of zero indicates perfect merit-proportional allocation, whereas larger values imply greater exposure imbalance. This group-sensitive evaluation is consistent with the motivation of Ma et al. [26], who emphasize the importance of accounting for heterogeneous data characteristics and systematic bias sources in advertising prediction models. Together with EFG, GCCEG provides a more comprehensive assessment of fairness in the proposed CGAD framework, as illustrated in Fig. 1.

IV. EXPERIMENTAL SETUP

A. Datasets and Semi-Synthetic Bias Injection

We use two public advertising click datasets: Criteo Display Advertising Challenge [17] and Avazu CTR Prediction [18]. For Criteo, we sample 5 million instances with an 8:1:1 train/validation/test split. For Avazu, we use the full dataset of approximately 40 million instances with the same split.

Since these are impression-level CTR datasets, we construct pseudo-query ranking lists by grouping impressions sharing contextual fields and sampling ten candidate ads per group. A proxy relevance model (a two-layer MLP) is trained on original click labels to produce synthetic relevance labels for evaluation.

We inject position bias with parameter η : the examination probability at position k is $P(o = 1|k) = (1/k)^\eta$. We define four ad groups via hashing selected categorical features and inject group bias with parameter γ : $P(o = 1|k, g) = (1/k)^\eta \cdot (1 - \gamma \cdot \mathbb{1}[g \in \mathcal{G}_{\text{disadv}}])$, where $\mathcal{G}_{\text{disadv}}$ contains two of the four groups, simulating reduced examination due to historical under-exposure.

B. Baselines and Implementation

We compare CGAD against five baselines: (1) **ERM**: empirical risk minimization on raw clicks; (2) **IPS**: standard inverse propensity scoring with position-only propensities; (3) **DLA**: dual learning algorithm with joint propensity-ranker estimation; (4) **FairReg**: ERM with a merit-proportional exposure regularizer; (5) **DR**: doubly robust estimation for position bias. All methods use the same two-layer MLP with 128 hidden units and ReLU activations, trained with Adam (lr 10^{-3}) for 50 epochs. For CGAD, we set $\lambda = 0.1$ by default.

C. Evaluation Metrics

We evaluate ranking accuracy using NDCG@10 and AUC computed against proxy relevance labels. For fairness, we report EFG (Eq. 6) and GCCEG (Eq. 8). We also report Worst-Group NDCG@10 (WG-NDCG). All results are averaged over five runs.

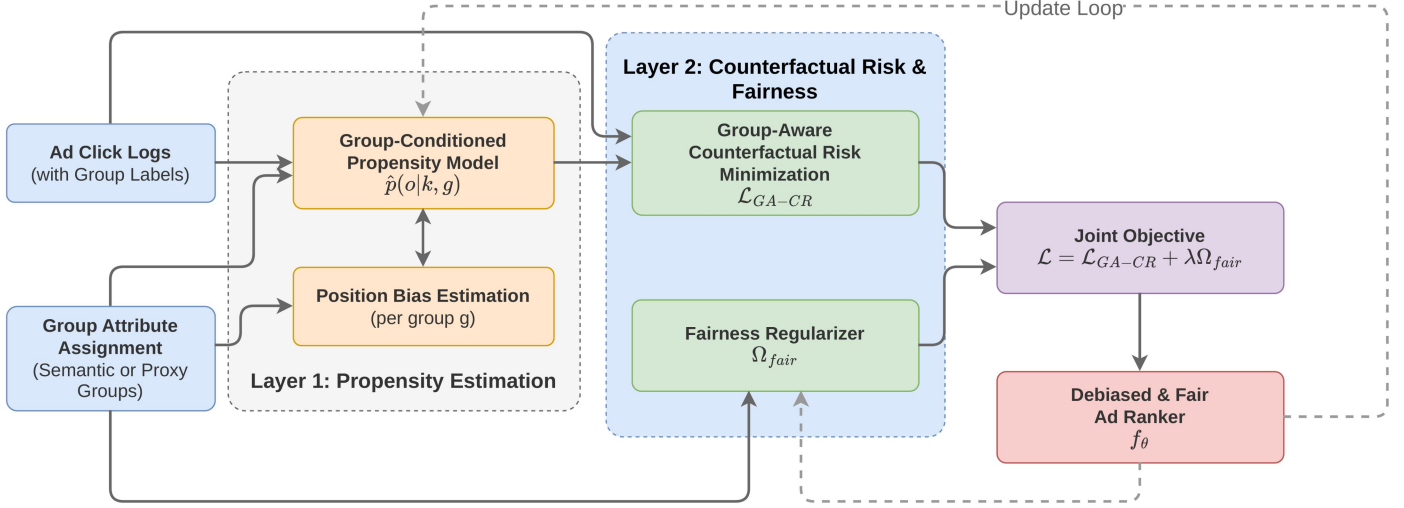


Fig. 1. Overview of the CGAD framework. Layer 1 estimates group-conditioned propensities $\hat{p}(o|k, g)$. Layer 2 combines $\mathcal{L}_{GA-CR}$ and Ω_{fair} into $\mathcal{L} = \mathcal{L}_{GA-CR} + \lambda \Omega_{fair}$ to produce a debiased and fair ranker f_{θ} . Dashed arrows indicate the training update loop.

V. RESULTS AND ANALYSIS

A. Main Results

Table I presents the results with $\eta = 0.6$ and $\gamma = 0.2$. CGAD achieves competitive ranking accuracy (NDCG@10 of 0.694 on Criteo, 0.681 on Avazu) while substantially reducing EFG by 46.5% relative to DLA on Criteo and 43.9% on Avazu. CGAD also achieves the lowest GCCEG, reducing it by 50.0% on Criteo and 48.4% on Avazu versus DLA. The WG-NDCG confirms that CGAD provides the most balanced performance across groups.

TABLE I.

MAIN RESULTS ($\eta = 0.6, \gamma = 0.2$). BEST IN **BOLD**, SECOND BEST UNDERLINED. $\downarrow$: LOWER IS BETTER. WG: WORST-GROUP NDCG@10.

Method	Criteo				
	NDCG@10	AUC	EFG $\downarrow$	GCCEG $\downarrow$	WG
ERM	0.651	0.762	0.312	0.194	0.581
IPS	0.678	0.781	0.279	0.168	0.612
DLA	0.685	0.789	0.258	0.152	0.624
DR	<u>0.691</u>	<u>0.794</u>	0.249	0.147	0.631
FairReg	0.662	0.773	<u>0.198</u>	<u>0.118</u>	0.628
CGAD	0.694	0.796	0.138	0.076	0.645
Method	Avazu				
	NDCG@10	AUC	EFG $\downarrow$	GCCEG $\downarrow$	WG
ERM	0.638	0.751	0.329	0.205	0.564
IPS	0.665	0.770	0.291	0.176	0.598
DLA	0.673	0.779	0.271	0.161	0.610
DR	<u>0.679</u>	<u>0.783</u>	0.262	0.155	0.618
FairReg	0.649	0.761	<u>0.209</u>	<u>0.124</u>	0.621
CGAD	0.681	0.786	0.152	0.083	0.641

B. Sensitivity to Bias Strength

Fig. 2 shows how NDCG@10 and EFG vary as η increases from 0.2 to 1.0 with $\gamma = 0.2$ on Criteo. As η grows, all methods experience degraded accuracy, but CGAD degrades more gracefully. More importantly, CGAD maintains a consistently lower EFG across all bias levels, with the advantage becoming more pronounced under stronger bias. At $\eta = 1.0$, CGAD

achieves an EFG of 0.231 compared to 0.385 for DLA and 0.328 for FairReg.

C. Accuracy-Fairness Trade-off and Worst-Group Analysis

Fig. 3 (left) presents a scatter plot of NDCG@10 versus EFG across 30 configurations (varying η and γ), each averaged over five runs. CGAD consistently occupies the desirable lower-right region (high accuracy, low disparity). Fig. 3 (right) shows a heatmap of Worst-Group NDCG@10 for CGAD across η and γ . Performance degrades smoothly with no abrupt failures; even at $\eta = 1.0, \gamma = 0.4$, Worst-Group NDCG remains at 0.553.

D. Ablation Study and Sensitivity to λ

Table II reports ablation results on Criteo ($\eta = 0.6, \gamma = 0.2$). Removing group-conditioned propensity (w/o GP) increases EFG from 0.138 to 0.211; removing the fairness regularizer (w/o FR) raises EFG to 0.197. Both components are essential.

Varying $\lambda \in \{0, 0.01, 0.05, 0.1, 0.2, 0.5\}$, EFG decreases steadily from 0.197 to 0.138 as λ increases from 0 to 0.1, while NDCG@10 slightly improves from 0.692 to 0.694. Beyond $\lambda = 0.2$, accuracy declines more noticeably. We use $\lambda = 0.1$ as the default.

TABLE II.

ABLATION STUDY ON CRITEO ($\eta = 0.6, \gamma = 0.2$). GP: GROUP-CONDITIONED PROPENSITY. FR: FAIRNESS REGULARIZATION.

Variant	NDCG@10	AUC	EFG $\downarrow$	GCCEG $\downarrow$	WG-NDCG
CGAD (full)	0.694	0.796	0.138	0.076	0.645
CGAD w/o GP	0.683	0.788	0.211	0.127	0.619
GP					
CGAD w/o FR	<u>0.692</u>	<u>0.793</u>	0.197	0.119	0.633
FR					

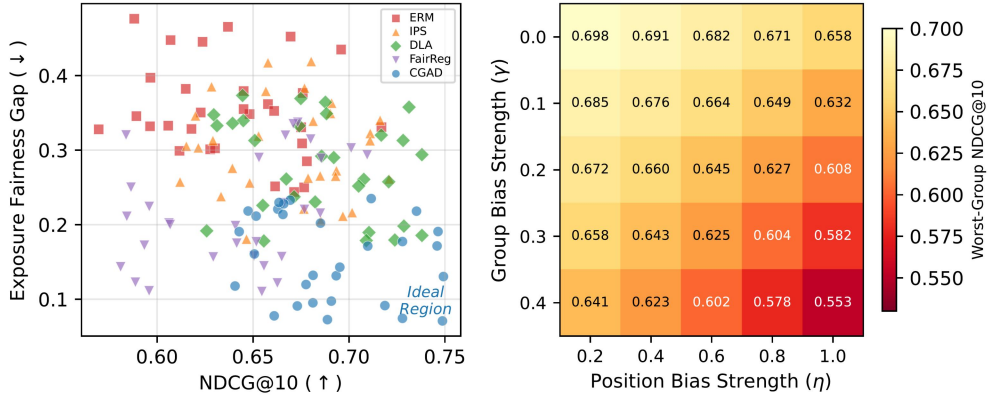


Fig. 2. NDCG@10 (left) and Exposure Fairness Gap (right) as functions of position bias strength η on Criteo ($\gamma = 0.2$). CGAD consistently achieves the best overall trade-off, with the advantage becoming more pronounced as position bias intensifies.

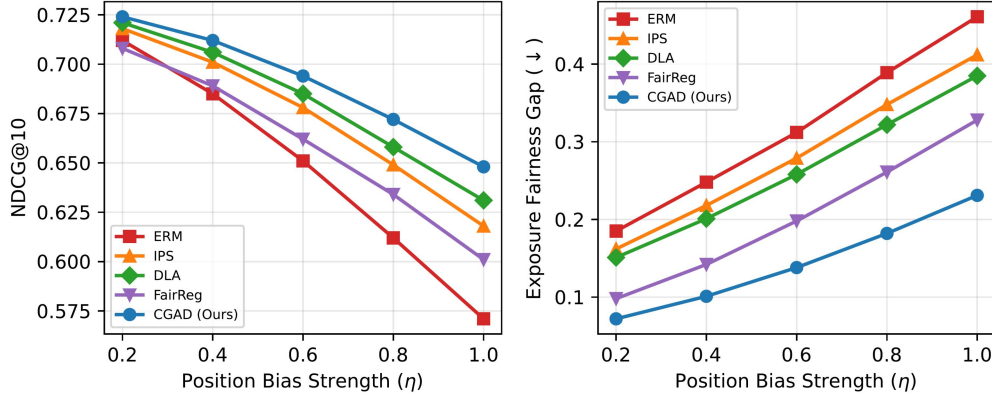


Fig. 3. (Left) Scatter plot of NDCG@10 vs. Exposure Fairness Gap across 30 configurations. CGAD consistently occupies the ideal lower-right region. (Right) Heatmap of Worst-Group NDCG@10 for CGAD across position bias strength η and group bias strength γ , showing graceful degradation.

VI. CONCLUSION

We have presented CGAD, a framework that jointly addresses position bias and group-level exposure confounding in ad ranking. Experiments on Criteo and Avazu with semi-synthetic bias injection show that CGAD achieves competitive ranking accuracy while reducing cross-group exposure disparity.

Limitations include reliance on semi-synthetic bias injection, proxy group definitions based on hashed features rather than semantic annotations, and the group-extended PBM assumption (Eq. 1). Future work could extend CGAD to more general click models and evaluate in online settings with naturally occurring group structures.

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